Q3. Bulbs </> Solved



## Problem Description

A wire connects **N** light bulbs.

Each bulb has a switch associated with it; however, due to faulty wiring, a switch also changes the state of all the bulbs to the right of the current bulb.

Given an initial state of all bulbs, find the **minimum number** of switches you have to press to turn on all the bulbs.

You can press the same switch multiple times.

**Note: 0** represents the bulb is off and **1** represents the bulb is on.

## Problem Constraints

$1 <= N <= 5\times10^5$
$0 <= A[i] <= 1$

## Input Format

The first and the only argument contains an integer array A, of size N.

## Output Format

Return an integer representing the minimum number of switches required.

## Example Input

Input 1:

```
A = [0, 1, 0, 1]
```

Input 2:

```
A = [1, 1, 1, 1]
```

## Example Output

Output 1:

```
4
```

Output 2:

```
0
```

## Example Explanation

Explanation 1:

```
press switch 0 : [1 0 1 0]
press switch 1 : [1 1 0 1]
press switch 2 : [1 1 1 0]
press switch 3 : [1 1 1 1]
```

Explanation 2:

```
There is no need to turn any switches as all the bulbs are already on.
```

See Expected Output

Java 8 (Oracle-Jdk-1.8)

```
/*
 * Time complexity = O(N) where 
 * */

import java.util.ArrayList;
import java.util.Scanner;

public class Solution
{
    public static void main(String[] a
    {
        Scanner sc = new Scanner(Syste

        String line1 = sc.nextLine();
        String[] strings = line1.split

        ArrayList<Integer> input = new

        for(String str : strings)
        {
            input.add(Integer.parseInt
        }

        System.out.print(bulbs(input))
```

Test Output

Test | Test With Custom Input | Submit